

線性代數 第二次小考複習：精選習題集（純題目）

範圍：Friedberg, Insel, Spence —— **Linear Algebra** (4th Edition), Chapter 3
Sect. 3.1、Sect. 3.2、Sect. 3.3、Sect. 3.4

📌 考試範圍與題目篩選說明（依夏良忠老師最新小考公告）：

1. 核心考點：第三章各節關於矩陣的 row echelon form、rank、inverse 之計算，以及線性方程組之求解。
2. 不考概念：排除 “dimension”、“linear independence” 等抽象向量空間概念。
3. 明確排除習題：
 - Sect. 3.2: 排除 6, 14
 - Sect. 3.3: 排除 1(h), 2, 6, 8, 9, 11 ~ 14
 - Sect. 3.4: 排除 1(f), 7 ~ 13
4. 本份題目集：已完整收錄上述範圍內的所有習題，保留原書精確英文題目與矩陣符號，排版清晰對標原版向量標準。

Section 3.1: Elementary Matrix Operations and Elementary Matrices

Exercise 1. Label the following statements as true or false.

- (a) An elementary matrix is always square.
- (b) The only entries of an elementary matrix are zeros and ones.
- (c) The $n \times n$ identity matrix is an elementary matrix.
- (d) The product of two $n \times n$ elementary matrices is an elementary matrix.
- (e) The inverse of an elementary matrix is an elementary matrix.
- (f) The sum of two $n \times n$ elementary matrices is an elementary matrix.
- (g) The transpose of an elementary matrix is an elementary matrix.
- (h) If B is a matrix that can be obtained by performing an elementary row operation on a matrix A , then B can also be obtained by performing an elementary column operation on A .
- (i) If B is a matrix that can be obtained by performing an elementary row operation on a matrix A , then A can be obtained by performing an elementary row operation on B .

Exercise 2. Let

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 0 & 1 \\ 1 & -1 & 1 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 0 & 3 \\ 1 & -2 & 1 \\ 1 & -3 & 1 \end{pmatrix}, \quad \text{and} \quad C = \begin{pmatrix} 1 & 0 & 3 \\ 0 & -2 & -2 \\ 1 & -3 & 1 \end{pmatrix}.$$

Find an elementary operation that transforms A into B and an elementary operation that transforms B into C . By means of several additional operations, transform C into I_3 .

Exercise 3. Use the proof of Theorem 3.2 to obtain the inverse of each of the following elementary matrices.

(a) $\begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix}$

(b) $\begin{pmatrix} 1 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 1 \end{pmatrix}$

(c) $\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -2 & 0 & 1 \end{pmatrix}$

Exercise 4. Prove the assertion made on page 149: Any elementary $n \times n$ matrix can be obtained in at least two ways—either by performing an elementary row operation on I_n or by performing an elementary column operation on I_n .

Exercise 5. Prove that E is an elementary matrix if and only if E^t is.

Exercise 6. Let A be an $m \times n$ matrix. Prove that if B can be obtained from A by an elementary row [column] operation, then B^t can be obtained from A^t by the corresponding elementary column [row] operation.

Exercise 7. Prove Theorem 3.1: Let $A \in M_{m \times n}(F)$, and suppose that B is obtained from A by performing an elementary row [column] operation. Then there exists an $m \times m$ [$n \times n$] elementary matrix E such that $B = EA$ [$B = AE$].

Exercise 8. Prove that if a matrix Q can be obtained from a matrix P by an elementary row operation, then P can be obtained from Q by an elementary row operation of the same type. (*Hint: Treat each type of elementary row operation separately.*)

Exercise 9. Prove that any elementary row [column] operation of type 1 can be obtained by a succession of three elementary row [column] operations of type 3 followed by one elementary row [column] operation of type 2.

Exercise 10. Prove that any elementary row [column] operation of type 2 can be obtained by dividing some row [column] by a nonzero scalar.

Exercise 11. Prove that any elementary row [column] operation of type 3 can be obtained by subtracting a multiple of some row [column] from another row [column].

Exercise 12. Let A be an $m \times n$ matrix. Prove that there exists a sequence of elementary row operations of types 1 and 3 that transforms A into an upper triangular matrix.

Section 3.2: The Rank of a Matrix and Matrix Inverses

(註: 依公告排除 Ex 6, 14; 關於向量空間、線性獨立之抽象證明不考)

Exercise 1. Label the following statements as true or false.

- (a) The rank of a matrix is equal to the number of its nonzero columns.
- (b) The product of two matrices always has rank equal to the lesser of the ranks of the two matrices.
- (c) The $m \times n$ zero matrix is the only $m \times n$ matrix having rank 0.
- (d) Elementary row operations preserve rank.
- (e) Elementary column operations do not necessarily preserve rank.
- (f) The rank of a matrix is equal to the maximum number of linearly independent rows in the matrix.
- (g) The inverse of a matrix can be computed exclusively by means of elementary row operations.
- (h) The rank of an $n \times n$ matrix is at most n .
- (i) An $n \times n$ matrix having rank n is invertible.

Exercise 2. Find the rank of the following matrices.

(a) $\begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \end{pmatrix}$

(b) $\begin{pmatrix} 1 & 1 & 0 \\ 2 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}$

(c) $\begin{pmatrix} 1 & 0 & 2 \\ 1 & 1 & 4 \end{pmatrix}$

(d) $\begin{pmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \end{pmatrix}$

(e) $\begin{pmatrix} 1 & 2 & 3 & 1 & 1 \\ 1 & 4 & 0 & 1 & 2 \\ 0 & 2 & -3 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 \end{pmatrix}$

(f) $\begin{pmatrix} 1 & 2 & 0 & 1 & 1 \\ 2 & 4 & 1 & 3 & 0 \\ 3 & 6 & 2 & 5 & 1 \\ -4 & -8 & 1 & -3 & 1 \end{pmatrix}$

(g) $\begin{pmatrix} 1 & 1 & 0 & 1 \\ 2 & 2 & 0 & 2 \\ 1 & 1 & 0 & 1 \\ 1 & 1 & 0 & 1 \end{pmatrix}$

Exercise 3. Prove that for any $m \times n$ matrix A , $\text{rank}(A) = 0$ if and only if A is the zero matrix.

Exercise 4. Use elementary row and column operations to transform each of the following matrices into a matrix D satisfying the conditions of Theorem 3.6 ($D = \begin{pmatrix} I_r & O \\ O & O \end{pmatrix}$), and then determine the rank of each matrix.

$$(a) \begin{pmatrix} 1 & 1 & 1 & 2 \\ 2 & 0 & -1 & 2 \\ 1 & 1 & 1 & 2 \end{pmatrix}$$

$$(b) \begin{pmatrix} 2 & 1 \\ -1 & 2 \\ 2 & 1 \end{pmatrix}$$

Exercise 5. For each of the following matrices, compute the rank and the inverse if it exists.

$$(a) \begin{pmatrix} 1 & 2 \\ 1 & 1 \end{pmatrix}$$

$$(b) \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}$$

$$(c) \begin{pmatrix} 1 & 2 & 1 \\ 1 & 3 & 4 \\ 2 & 3 & -1 \end{pmatrix}$$

$$(d) \begin{pmatrix} 0 & -2 & 4 \\ 1 & 1 & -1 \\ 2 & 4 & -5 \end{pmatrix}$$

$$(e) \begin{pmatrix} 1 & 2 & 1 \\ -1 & 1 & 2 \\ 1 & 0 & 1 \end{pmatrix}$$

$$(f) \begin{pmatrix} 1 & 2 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 1 \end{pmatrix}$$

$$(g) \begin{pmatrix} 1 & 2 & 1 & 0 \\ 2 & 5 & 5 & 1 \\ -2 & -3 & 0 & 3 \\ 3 & 4 & -2 & -3 \end{pmatrix}$$

$$(h) \begin{pmatrix} 1 & 0 & 1 & 1 \\ 1 & 1 & -1 & 2 \\ 2 & 0 & 1 & 0 \\ 0 & -1 & 1 & -3 \end{pmatrix}$$

Exercise 7. Express the invertible matrix

$$\begin{pmatrix} 1 & 2 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 2 \end{pmatrix}$$

as a product of elementary matrices.

Exercise 8. Let A be an $m \times n$ matrix. Prove that if c is any nonzero scalar, then $\text{rank}(cA) = \text{rank}(A)$.

Exercise 11. Let

$$B = \begin{pmatrix} 1 & 0 & \dots & 0 \\ 0 \\ \vdots & B' \\ 0 \end{pmatrix}$$

where B' is an $m \times n$ submatrix of B . Prove that if $\text{rank}(B) = r$, then $\text{rank}(B') = r - 1$.

Exercise 12. Let B' and D' be $m \times n$ matrices, and let B and D be $(m+1) \times (n+1)$ matrices respectively defined by

$$B = \begin{pmatrix} 1 & 0 & \dots & 0 \\ 0 \\ \vdots & B' \\ 0 \end{pmatrix} \quad \text{and} \quad D = \begin{pmatrix} 1 & 0 & \dots & 0 \\ 0 \\ \vdots & D' \\ 0 \end{pmatrix}.$$

Prove that if B' can be transformed into D' by an elementary row [column] operation, then B can be transformed into D by an elementary row [column] operation.

Exercise 15. Suppose that A and B are matrices having n rows. Prove that $M(A \mid B) = (MA \mid MB)$ for any $m \times n$ matrix M .

Exercise 17. Prove that if B is a 3×1 matrix and C is a 1×3 matrix, then the 3×3 matrix BC has rank at most 1. Conversely, show that if A is any 3×3 matrix having rank 1, then there exist a 3×1 matrix B and a 1×3 matrix C such that $A = BC$.

Exercise 18. Let A be an $m \times n$ matrix and B be an $n \times p$ matrix. Prove that AB can be written as a sum of n matrices of rank at most one.

Exercise 19. Let A be an $m \times n$ matrix with rank m and B be an $n \times p$ matrix with rank n . Determine the rank of AB . Justify your answer.

Exercise 20. Let

$$A = \begin{pmatrix} 1 & 0 & -1 & 2 & 1 \\ -1 & 1 & 3 & -1 & 0 \\ -2 & 1 & 4 & -1 & 3 \\ 3 & -1 & -5 & 1 & -6 \end{pmatrix}.$$

(a) Find a 5×5 matrix M with rank 2 such that $AM = O$, where O is the 4×5 zero matrix.

(b) Suppose that B is a 5×5 matrix such that $AB = O$. Prove that $\text{rank}(B) \leq 2$.

Exercise 21. Let A be an $m \times n$ matrix with rank m . Prove that there exists an $n \times m$ matrix B such that $AB = I_m$.

Exercise 22. Let B be an $n \times m$ matrix with rank m . Prove that there exists an $m \times n$ matrix A such that $AB = I_m$.

Section 3.3: Systems of Linear Equations—Theoretical Aspects

(註:依公告排除 Ex 1(h), 2, 6, 8, 9, 11 ~ 14)

Exercise 1. Label the following statements as true or false.

- (a) Any system of linear equations has at least one solution.
- (b) Any system of linear equations has at most one solution.
- (c) Any homogeneous system of linear equations has at least one solution.
- (d) Any system of n linear equations in n unknowns has at most one solution.
- (e) Any system of n linear equations in n unknowns has at least one solution.
- (f) If the homogeneous system corresponding to a given system of linear equations has a solution, then the given system has a solution.
- (g) If the coefficient matrix of a homogeneous system of n linear equations in n unknowns is invertible, then the system has no nonzero solutions.
- (h) [排除] The solution set of any system of m linear equations in n unknowns is a subspace of F^n .

Exercise 3. Find all solutions to the following systems of linear equations.

- (a) $\begin{cases} x_1 + 3x_2 = 5 \\ 2x_1 + 6x_2 = 10 \end{cases}$
- (b) $\begin{cases} x_1 + x_2 - x_3 = 1 \\ 4x_1 + x_2 - 2x_3 = 3 \end{cases}$
- (c) $\begin{cases} x_1 + 2x_2 - x_3 = 3 \\ 2x_1 + x_2 + x_3 = 6 \end{cases}$
- (d) $\begin{cases} 2x_1 + x_2 - x_3 = 5 \\ x_1 - x_2 + x_3 = 1 \\ x_1 + 2x_2 - 2x_3 = 4 \end{cases}$
- (e) $x_1 + 2x_2 - 3x_3 + x_4 = 1$
- (f) $\begin{cases} x_1 + 2x_2 = 5 \\ x_1 - x_2 = -1 \end{cases}$
- (g) $\begin{cases} x_1 + 2x_2 + x_3 + x_4 = 1 \\ x_2 - x_3 + x_4 = 1 \end{cases}$

Exercise 4. For each system of linear equations with the invertible coefficient matrix A , (1) Compute A^{-1} , and (2) Use A^{-1} to solve the system.

$$(a) \begin{cases} x_1 + 3x_2 = 4 \\ 2x_1 + 5x_2 = 3 \end{cases}$$

$$(b) \begin{cases} x_1 + 2x_2 - x_3 = 5 \\ x_1 + x_2 + x_3 = 1 \\ 2x_1 - 2x_2 + x_3 = 4 \end{cases}$$

Exercise 5. Give an example of a system of n linear equations in n unknowns with infinitely many solutions.

Exercise 7. Determine which of the following systems of linear equations has a solution.

$$(a) \begin{cases} x_1 + x_2 - x_3 + 2x_4 = 2 \\ x_1 + x_2 + 2x_3 = 1 \\ 2x_1 + 2x_2 + x_3 + 2x_4 = 4 \end{cases}$$

$$(b) \begin{cases} x_1 + x_2 - x_3 = 1 \\ 2x_1 + x_2 + 3x_3 = 2 \end{cases}$$

$$(c) \begin{cases} x_1 + 2x_2 + 3x_3 = 1 \\ x_1 + x_2 - x_3 = 0 \\ x_1 + 2x_2 + x_3 = 3 \end{cases}$$

$$(d) \begin{cases} x_1 + x_2 + 3x_3 - x_4 = 0 \\ x_1 + x_2 + x_3 + x_4 = 1 \\ x_1 - 2x_2 + x_3 - x_4 = 1 \\ 4x_1 + x_2 + 8x_3 - x_4 = 0 \end{cases}$$

$$(e) \begin{cases} x_1 + 2x_2 - x_3 = 1 \\ 2x_1 + x_2 + 2x_3 = 3 \\ x_1 - 4x_2 + 7x_3 = 4 \end{cases}$$

Exercise 10. Prove or give a counterexample to the following statement: If the coefficient matrix of a system of m linear equations in n unknowns has rank m , then the system has a solution.

Section 3.4: Systems of Linear Equations—Computational Aspects

(註:依公告排除 Ex 1(f), 7 ~ 13)

Exercise 1. Label the following statements as true or false.

- (a) If $(A' \mid b')$ is obtained from $(A \mid b)$ by a finite sequence of elementary column operations, then the systems $Ax = b$ and $A'x = b'$ are equivalent.
- (b) If $(A' \mid b')$ is obtained from $(A \mid b)$ by a finite sequence of elementary row operations, then the systems $Ax = b$ and $A'x = b'$ are equivalent.
- (c) If A is an $n \times n$ matrix with rank n , then the reduced row echelon form of A is I_n .
- (d) Any matrix can be put in reduced row echelon form by means of a finite sequence of elementary row operations.
- (e) If $(A \mid b)$ is in reduced row echelon form, then the system $Ax = b$ is consistent.
- (f) [排除] Let $Ax = b$ be a system of m linear equations in n unknowns for which the augmented matrix is in reduced row echelon form...
- (g) If a matrix A is transformed by elementary row operations into a matrix A' in reduced row echelon form, then the number of nonzero rows in A' equals the rank of A .

Exercise 2. Use Gaussian elimination to solve the following systems of linear equations.

$$(a) \begin{cases} x_1 + 2x_2 - x_3 = -1 \\ 2x_1 + 2x_2 + x_3 = 1 \\ 3x_1 + 5x_2 - 2x_3 = -1 \end{cases}$$

$$(b) \begin{cases} x_1 - 2x_2 - x_3 = 1 \\ 2x_1 - 3x_2 + x_3 = 6 \\ 3x_1 - 5x_2 = 7 \\ x_1 + 5x_3 = 9 \end{cases}$$

$$(c) \begin{cases} x_1 + 2x_2 + 2x_4 = 6 \\ 3x_1 + 5x_2 - x_3 + 6x_4 = 17 \\ 2x_1 + 4x_2 + x_3 + 2x_4 = 12 \\ 2x_1 - 7x_3 + 11x_4 = 7 \end{cases}$$

$$(d) \begin{cases} x_1 - x_2 - 2x_3 + 3x_4 = -7 \\ 2x_1 - x_2 + 6x_3 + 6x_4 = -2 \\ -2x_1 + x_2 - 4x_3 - 3x_4 = 0 \\ 3x_1 - 2x_2 + 9x_3 + 10x_4 = -5 \end{cases}$$

$$(e) \begin{cases} x_1 - 4x_2 - x_3 + x_4 = 3 \\ 2x_1 - 8x_2 + x_3 - 4x_4 = 9 \\ -x_1 + 4x_2 - 2x_3 + 5x_4 = -6 \end{cases}$$

$$(f) \begin{cases} x_1 + 2x_2 - x_3 + 3x_4 = 2 \\ 2x_1 + 4x_2 - x_3 + 6x_4 = 5 \\ x_2 + 2x_4 = 3 \end{cases}$$

$$(g) \begin{cases} 2x_1 - 2x_2 - x_3 + 6x_4 - 2x_5 = 1 \\ x_1 - x_2 + x_3 + 2x_4 - x_5 = 2 \\ 4x_1 - 4x_2 + 5x_3 + 7x_4 - x_5 = 6 \end{cases}$$

$$\begin{aligned}
\text{(h)} \quad & \begin{cases} 3x_1 - x_2 + x_3 - x_4 + 2x_5 = 5 \\ x_1 - x_2 - x_3 - 2x_4 - x_5 = 2 \\ 5x_1 - 2x_2 + x_3 - 3x_4 + 3x_5 = 10 \\ 2x_1 - x_2 - 2x_4 + x_5 = 5 \end{cases} \\
\text{(i)} \quad & \begin{cases} 3x_1 - x_2 + 2x_3 + 4x_4 + x_5 = 2 \\ x_1 - x_2 + 2x_3 + 3x_4 + x_5 = -1 \\ 2x_1 - 3x_2 + 6x_3 + 9x_4 + 4x_5 = -5 \\ 7x_1 - 2x_2 + 4x_3 + 8x_4 + x_5 = 6 \end{cases} \\
\text{(j)} \quad & \begin{cases} 2x_1 + 3x_3 - 4x_5 = 5 \\ 3x_1 - 4x_2 + 8x_3 + 3x_4 = 8 \\ x_1 - x_2 + 2x_3 + x_4 - x_5 = 2 \\ -2x_1 + 5x_2 - 9x_3 - 3x_4 - 5x_5 = -8 \end{cases}
\end{aligned}$$

Exercise 3. Suppose that the augmented matrix of a system $Ax = b$ is transformed into a matrix $(A' \mid b')$ in reduced row echelon form by a finite sequence of elementary row operations. (a) Prove that $\text{rank}(A') \neq \text{rank}(A' \mid b')$ if and only if $(A' \mid b')$ contains a row in which the only nonzero entry lies in the last column. (b) Deduce that $Ax = b$ is consistent if and only if $(A' \mid b')$ contains no row in which the only nonzero entry lies in the last column.

Exercise 4. For each of the systems that follow, apply Exercise 3 to determine whether the system is consistent. If the system is consistent, find all solutions.

$$\begin{aligned}
\text{(a)} \quad & \begin{cases} x_1 + 2x_2 - x_3 + x_4 = 2 \\ 2x_1 + x_2 + x_3 - x_4 = 3 \\ x_1 + 2x_2 - 3x_3 + 2x_4 = 2 \end{cases} & \text{(b)} \quad \begin{cases} x_1 + x_2 - 3x_3 + x_4 = -2 \\ x_1 + x_2 + x_3 - x_4 = 2 \\ x_1 + x_2 - x_3 = 0 \end{cases} \\
\text{(c)} \quad & \begin{cases} x_1 + x_2 - 3x_3 + x_4 = 1 \\ x_1 + x_2 + x_3 - x_4 = 2 \\ x_1 + x_2 - x_3 = 0 \end{cases}
\end{aligned}$$

Exercise 5. Let the reduced row echelon form of A be

$$\begin{pmatrix} 1 & 0 & 2 & 0 & -2 \\ 0 & 1 & -5 & 0 & -3 \\ 0 & 0 & 0 & 1 & 6 \end{pmatrix}.$$

Determine A if the first, second, and fourth columns of A are

$$\begin{pmatrix} 1 \\ -1 \\ 3 \end{pmatrix}, \quad \begin{pmatrix} 0 \\ -1 \\ 1 \end{pmatrix}, \quad \text{and} \quad \begin{pmatrix} 1 \\ -2 \\ 0 \end{pmatrix},$$

respectively.

Exercise 6. Let the reduced row echelon form of A be

$$\begin{pmatrix} 1 & -3 & 0 & 4 & 0 & 5 \\ 0 & 0 & 1 & 3 & 0 & 2 \\ 0 & 0 & 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}.$$

Determine A if the first, third, and fifth columns of A are

$$\begin{pmatrix} 1 \\ 2 \\ 3 \\ -1 \end{pmatrix}, \quad \begin{pmatrix} 2 \\ -1 \\ 1 \\ 0 \end{pmatrix}, \quad \text{and} \quad \begin{pmatrix} 1 \\ 0 \\ 2 \\ 2 \end{pmatrix},$$

respectively.